

Classwork 01// Matrix Calculus: Jacobian

Student Name: _____

Points: 30

A Number: _____

1. Derive the Jacobian of identity function $\mathbf{f}(\mathbf{x}) = \mathbf{x}$, i.e. $f_i(\mathbf{x}) = x_i$ that has n functions and each function has n parameters held in a single vector $\mathbf{x}$. **(10 Points)**
2. Under what conditions, are the off-diagonal elements of a Jacobian zero? **(10 Points)**
3. Compute gradient of $y = \text{sum}(\mathbf{x})$, i.e. ∇y , considering the function inside the summation as $f_i(\mathbf{x}) = x_i$. Is it a vertical vector or a horizontal vector as per convention used in the Lecture notes? **(10 Points)**